

How to Study This Course

A Step-by-Step Study Guide — Clinical Data Analysis in R (Phase I)

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Not sure where to begin? You are in the right place. This guide takes you, one small step at a time, from a blank laptop to being ready for the first session. You do **not** need any programming experience — just follow the steps in order. Most take only a few minutes.

1. How the course is organised

The course lives in a few clearly named folders. This week you only need the first one.

Folder	What it is / when to use it
Course_Preparation/	START HERE. Your self-study pack for the week before the course. Everything in this guide happens here.
Course/	The main course used during the live sessions (lecture materials, scripts, data). You do not need it yet.
Deck_and_Build/	The slide decks and the tools that build them (for the trainers). You can ignore this.

Inside **Course_Preparation/** the folders are numbered in the exact order you should follow them:

Folder	What you do there
00_READ_ME_FIRST/	Read the welcome and this Study Guide.
01_Install_R_and_RStudio/	Install the software (Steps 2–4 below).
02_Getting_Started_with_RStudio/	A short tour of the RStudio screen.
03_R_Basics/ → 07_Regression/	The five lessons, in order.
08_Advanced_Model/	Optional bonus lesson (survival analysis).
Data/	The practice dataset and its data dictionary.
Exercises/ and Solutions/	Short practice tasks and their answers.
Cheat_Sheets/ and Guides/	Quick references to keep open while you work.

2. Install R

R is the free engine that does the analysis. Install it first.

1. Go to <https://cran.r-project.org>
2. **Windows:** click *Download R for Windows* → *base* → *Download R-4.x.x for Windows*, then run the installer and accept the defaults.
3. **macOS:** click *Download R for macOS* and choose the package for your Mac (the **-arm64** one for Apple M1/M2/M3 chips; the Intel one for older Macs). Open the .pkg and follow the steps.

3. Install RStudio Desktop

RStudio is the friendly window you actually work in. Install R first, then RStudio.

1. Go to <https://posit.co/download/rstudio-desktop/>
2. Download the free **RStudio Desktop** for your system (Windows .exe or macOS .dmg) and install it.
3. Open **RStudio** (not plain R). RStudio runs R for you.

4. Install the R packages

Packages are free add-ons that give R extra tools. Open RStudio, click in the **Console** (the bottom-left pane), paste the line below, and press Enter. It may take a few minutes.

```
install.packages(c("tidyverse", "readxl", "gtsummary",  
                  "broom", "survival", "survminer"))
```

TIP You only install a package once per computer. In every new session you simply load it with `library(...)` — for example `library(tidyverse)`.

5. Check that it all works

In the **01_Install_R_and_RStudio/** folder, open `installation_test.R` in RStudio and run it (click **Source**, top-right of the editor). If everything is installed you will see:

```
My R and RStudio installation is working.
```

You can also type these four lines in the Console to check R itself:

```
2 + 2  
x <- c(10, 12, 14, 16)  
mean(x)  
library(dplyr)
```

STUCK? See the troubleshooting section in `Install_R_and_RStudio.md`, or email My Luong Vuong at myluong1710@gmail.com. It is fine to arrive with setup half-working — there will be help on the first evening.

6. Open the course project

The pre-course pack is an **RStudio Project**. Opening it points RStudio at the right folder so every file path in the lessons just works.

1. Open the **Course_Preparation** folder on your computer.
2. Double-click `Clinical_Data_Analysis_PreCourse.Rproj`. RStudio opens with the project loaded.
3. You will see the project name at the top-right of RStudio. From now on, always open the course this way.

WHY IT MATTERS Because you opened the project, a path like `read_csv("Data/clinical_data_clean.csv")` finds the data automatically — you never have to edit a file path.

7. How to open and run the lessons

The lessons are files you open inside RStudio. There are two kinds:

- **.R files** (scripts) — plain R code. The exercises and solutions are .R files.
- **.Rmd files** (R Markdown) — explanation and code together. The six lessons are .Rmd files.

Open a file

Use the **Files** pane (bottom-right) to browse to a folder, then click a file to open it in the editor (top-left).

Run one line

Click on a line of code, then press **Ctrl+Enter** (Windows) or **Cmd+Enter** (Mac). The result appears in the Console below.

Run several lines

Select the lines with your mouse, then press **Ctrl+Enter** / **Cmd+Enter**. They run in order.

See a whole lesson with its output (Knit)

For an .Rmd lesson, click the **Knit** button at the top of the editor. RStudio runs everything and produces a tidy document with the code, its output, and the figures. This is the easiest way to read a lesson.

IF CODE SEEMS STUCK A red Stop sign appears in the Console while code runs. Click it (or press Esc in the Console) to stop.

8. Your study path for the week

Work through the pack in this order — about 2–3 hours in total, spread over a few short evenings.

1. Install and test (Steps 2–5 above).
2. Open the project (Step 6) and read **02_Getting_Started_with_RStudio**.
3. Do the lessons in order: **03_R_Basics** → **04_Data_Management** → **05_Exploratory_Data_Analysis** → **06_Statistical_Tests** → **07_Regression**. (08 is optional.)
4. After a lesson, try the matching exercise in **Exercises/**, then check **Solutions/**.
5. Keep **Cheat_Sheets/** and **Guides/** open for quick reference.

HOW TO STUDY WELL Type the code yourself instead of copying — it sticks better. Change a number and re-run to see what happens; you cannot break anything. Nothing here is graded; the goal is confidence.

You are ready when...

- R and RStudio are installed and `installation_test.R` prints the success message.

- You can open the .Rproj and load the dataset with read_csv.
- You can run one line and several lines of code.
- You have looked at R Basics and Data Management.

See you on **8 September**. — Bernard & My Luong, Neudata · #ClearDataClearImpact